APPENDIX

This appendix lists all the postulates for plane geometry, along with some of the definitions and theorems. In addition to being convenient for reference purposes, the appendix merits study, because reading it over and thinking about the order of the theorems will help you fix the structure of Euclidean geometry in your mind.

THE POSTULATES

There are three groups of postulates, and three postulates listed singly. They are:

Postulates of incidence (3 postulates)
Postulates of order or betweenness (5 postulates)
Postulates of congruence (7 postulates)
The postulate of Archimedes
The completeness postulate
The parallel postulate.

The Incidence Postulates

1. There are at least three points not all on a line.
2. For any two different points, there is exactly one line containing these points.
3. Every line contains at least two points.

The Betweenness Postulates

4. If B is between A and C, then A, B, and C are three different, or distinct, points on a line.
5. For every three points on a line, exactly one of them is between the other two.
6. Any four points on a line may be named A_1, A_2, A_3, A_4, so that the only betweenness relations are the same as the order of the subscript numbers. That is, the betweenness relations are: A_2 between A_1 and A_3; A_2 between A_1 and A_4; etc.
7. If A and B are two points, then there is at least one point C such that B is between A and C, and at least one point D such that D is between A and B.
8. Every line separates the plane. By this we mean that all points of the plane not on the line are divided into two sets, called the two sides of the line, having the following properties:
   (a) If P and Q belong to one of these sets, then no point of l is between P and Q. We say then that P and Q are on the same side of l.
   (b) If P and Q are in different sets, then there is a point on l which is between P and Q.

Linear Congruence Postulates

9. Given points A and B on a line l and a point A' on a line l', then on a given side of A' on l' there is exactly one point B' such that AB is congruent to $A'B'$. To indicate that AB is congruent to $A'B'$, we write "$AB \cong A'B'$."
   In congruence the order of the points does not matter, that is, if $AB \cong A'B'$, then also $AB \cong B'A'$, $BA \cong A'B'$, and $BA \cong B'A'$.
10. Congruence also satisfies the following:
    (a) $AB \cong AB$.
    (b) If $AB \cong A'B'$, then $A'B' \cong AB$.
    (c) If $AB \cong A'B'$ and $A'B' \cong A''B''$, then $AB \cong A''B''$.
11. (This postulate tells how to "add" and "subtract" segments.) Suppose that B is between A and C on a line l, and that B' is between A' and C' on a line l'.
    (a) If $AB \cong A'B'$ and $BC \cong B'C'$, then $AC \cong A'C'$.
    (b) If $AC \cong A'C'$ and $BC \cong B'C'$, then $AB \cong A'B'$.

Archimedes' Postulate

12. If AB is any segment on a line l and CD any other segment, then there is a finite number n of points $A_1, A_2, \ldots, A_n$ on l such that the points $A, A_1, A_2, \ldots, A_n$ are arranged in this order, and all the segments $AA_1, A_1A_2, \ldots, A_{n-1}A_n$ are congruent to CD, and either $B = A_n$ or B lies between A_{n-1} and A_n.

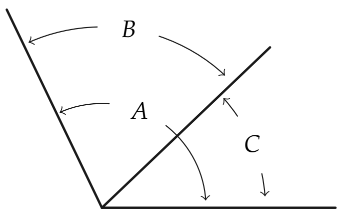
The Completeness Postulate

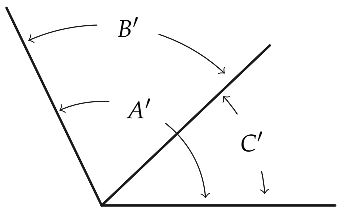
13. For every line l and any given point A on l, and for any positive real number x, there is, on a given side of A, a point B such that $\overline{AB} = x$.

Congruence Postulates for Angles

14. If $\angle A$ is not a straight angle and if r' is a ray from A' on a line l', then on a given side of l' there is exactly one ray s' from A' such that the angle A', with sides r' and s', is congruent to angle A. We write this as "$\angle A' \cong \angle A$." If $\angle A$ is a straight angle, then the straight angle at A' with one side r' is the only angle with one side r' congruent to angle A.
15.
   (a) $\angle A \cong \angle A$.
   (b) If $\angle A \cong \angle B$, then $\angle B \cong \angle A$.
   (c) If $\angle A \cong \angle B$ and $\angle B \cong \angle C$, then $\angle A \cong \angle C$.
16. If in the figure any two of the pairs $\angle A$ and $\angle A'$, $\angle B$ and $\angle B'$, $\angle C$ and $\angle C'$ are congruent, then the third pair of angles are congruent. In other words:
   (a) If $\angle A \cong \angle A'$ and $\angle B \cong \angle B'$, then $\angle C \cong \angle C'$.
   (b) If $\angle B \cong \angle B'$ and $\angle C \cong \angle C'$, then $\angle A \cong \angle A'$.
   (c) If $\angle C \cong \angle C'$ and $\angle A \cong \angle A'$, then $\angle B \cong \angle B'$.

   The angle A may be a straight angle.





17. If for $\triangle ABC$ and $\triangle A'B'C'$, $AB \cong A'B'$, $AC \cong A'C'$, and $\angle A \cong \angle A'$, then $\angle B \cong \angle B'$ and $\angle C \cong \angle C'$.

The Parallel Postulate

18. Through a point not on a line there is no more than one line parallel to the given line.

SELECTED DEFINITIONS

Although every definition has a purpose, certain ones are more necessary than others, and most of these essential definitions are listed here. In some cases an abbreviated form of the definition is given when no misunderstanding can arise.

1. Each set of points in the plane is called a *geometric figure*.
2. If A and B are any two points on a line, then the set of points consisting of A and B and all the points between A and B will be called the *line segment* (or *segment*) determined by A and B. The segment will be indicated by "AB." The points A and B are called the *ends* of the segment.
3. If A, B, and C are three points not all on a line, the set of points of the segments AB, BC, AC is called a *triangle*. We call the points A, B, and C *vertices* and the segments AB, AC, BC *sides* of the triangle. To indicate "triangle ABC," we write "$\triangle ABC$."
4. If O is a point of a line l and A and B are two other points of l, then we say that A and B are on the *same side of* O if O is not between A and B. We say that A and B are on *opposite sides of* O if O is between A and B.
5. A *ray* from O, or a ray with end O, is a set of points consisting of O and all the points on one side of O of a line l through O.

6. The set of all points on two rays from a point O is called an *angle*. The point O is called the *vertex* of the angle, and the two rays are called the *sides* of the angle. If the two rays are on the same line, the angle is called a *straight angle*.
7. Consider an angle, not a straight angle, with vertex O and whose sides are the rays r_1 and r_2 on the lines l_1 and l_2, respectively. The *interior of the angle* is the set of all points on the same side of l_1 as r_2 and on the same side of l_2 as r_1. The *exterior of the angle* is the set of all points in the plane which are not points of the angle and are not in the interior of the angle.
8. Two angles are *adjacent* to each other if they have a common side and vertex and have no common interior points.
9. Let AB be any segment and $A'B'$ any other segment on a line l'. On the same side of A' as B', let B'' be the point such that $AB \cong A'B''$. If B' is between A' and B'', we say that $AB > A'B'$. If B'' is between A' and B', we say that $AB < A'B'$.
10. For the definition of *length of a segment* read Chapter 5, Section 4.
11. A *right* angle is one which is congruent to its supplement.
12. Two triangles are said to be *congruent* if they can be labeled $\triangle ABC$ and $\triangle A'B'C'$ such that $AB \cong A'B'$, $AC \cong A'C'$, $BC \cong B'C'$, $\angle A \cong \angle A'$, $\angle B \cong \angle B'$, and $\angle C \cong \angle C'$. To indicate that "triangle ABC is congruent to triangle $A'B'C'$," we write "$\triangle ABC \cong \triangle A'B'C'$."
13. Let A and B be any angles, not straight angles. Let $\angle B'$ be an angle congruent to $\angle B$ such that one side of $\angle B'$ is the same as the first side of $\angle A$, and the other side of $\angle B'$ and the second side of $\angle A$ lie on the same side of the first side of $\angle A$. If the second side of $\angle B'$ is interior to $\angle A$, we say $\angle B < \angle A$. If the second side of $\angle B'$ is exterior to $\angle A$, we say $\angle B > \angle A$. If $\angle A$ is a straight angle and $\angle B$ is not a straight

angle, we say $\angle A > \angle B$ and $\angle B < \angle A$.

14. Let $P_1, P_2, \ldots, P_n$ be any set of n points. The set of points on all the segments $P_1P_2, P_2P_3, \ldots, P_{n-1}P_n$ is called a *polygonal path*. The path is said to join, or connect, the endpoints P_1 and P_n. The points $P_1, \ldots, P_n$ are called the *vertices*. In particular, if $P_1 = P_n$, the polygonal path is called a *polygon* and designated as polygon $P_1P_2 \ldots P_{n-1}$.
    A polygon is *convex* if for every two points on the polygon each point between them is either on the polygon or in the interior. A convex, equiangular, and equilateral polygon is called *regular*.
15. Two lines are *parallel* if they do not intersect.
16. If the ray OB is in the interior of $\angle AOC$ (or $\angle AOC$ is straight), then $\angle AOC$ is the *sum* of $\angle AOB$ and $\angle BOC$.
17. If AB and CD are segments, the *ratio* of AB to CD is the number $\overline{AB} \div \overline{CD}$. The statement that the segments AB and CD are *proportional* to the segments $A'B'$ and $C'D'$ means that $\overline{AB}/\overline{A'B'} = \overline{CD}/\overline{C'D'}$. We say that the sides of $\triangle ABC$ are proportional to the sides of $\triangle A'B'C'$ if $\overline{AB}/\overline{A'B'} = \overline{BC}/\overline{B'C'} = \overline{CA}/\overline{C'A'}$.
18. A *circle* is determined by a point O, called the center, and a positive number r. The circle is the set of all points A of the plane such that the distance $\overline{OA} = r$. If A is a point of the circle, then the segment OA is called a *radius* (plural, *radii*). If A and B are on the circle and the segment AB contains O, then AB is called a *diameter*.
19. If a circle has center O and radius r, the *interior of the circle* is the set of all points B such that $\overline{OB} < r$. The *exterior of the circle* is the set of all points C such that $\overline{OC} > r$. A *chord* is a segment determined by two points E and F on the circle. The line through E and F is a *secant*. A *central angle* of a circle is an angle whose vertex is at the center of the circle. An *inscribed*

angle is an angle with vertex on the circle and a second point of the circle on each of the two sides of the angle.

20. Suppose that for each positive integer n there is associated in some way, a number x_n. The numbers $x_1, x_2, \ldots, x_n$ then form an *infinite sequence*. The number x_n is called the nth term of the sequence. We designate the sequence by the symbol $\{x_n\}$.
21. The number L is called the *limit of the sequence* x_n if, for large values of n, x_n is necessarily arbitrarily close to L. We then say that L is the limit of the sequence of terms x_n and write $L = \lim\{x_n\}$.
22. The *circumference* C of a circle is the limit of the perimeters p_n of regular inscribed polygons. That is, $C = \lim\{p_n\}$.
23. The number C/d, which is the same for all circles, will be denoted by the Greek letter π (pi).
24. The *area* S *of a circle* is the limit of the areas of inscribed regular polygons. That is, $S = \lim\{S_n\}$.
25. Given a circular arc, let l_n be the length of a regular polygonal path of 2^n sides inscribed in the arc. The *length* l *of the arc* is defined to be the following limit: $l = \text{limit}\,\{l_n\}$.
26. On a circle of radius r an arc of length $\pi r/180$ is subtended by a central angle of *one degree* (1°).
27. If on a circle of radius r a central angle subtends an arc of length L, then the *measure of the angle in degrees* is $(L/\pi r) \cdot 180^\circ$.

SELECTED THEOREMS

The theorems selected below have been chosen to help you remember how the ideas have been developed and *in what order*.

There is no attempt at completeness, and in some cases the statement is less precise than in the text; when this is the case, you should try to fill in the details or refer back to the text.

1. Two different lines intersect in at most one point.
2. Let O be any point of a line l. The point O separates the line into two sets, called the two sides of the point O on the line, having the following properties: (1) If P and Q belong to the same side of O, then either OPQ or OQP. (2) If P and Q belong to different sides of O, then POQ.
3. If P and Q are both in the interior of $\angle O$, then the segment PQ is in the interior of $\angle O$.
4. If rays r_1 and r_2 form $\angle O$, a ray r from O, other than r_1 or r_2 lies either completely in the interior or completely in the exterior of $\angle O$.
5. If P is in the interior of $\angle O$ and Q is in the exterior of $\angle O$, then the segment PQ contains a point of the angle.
6. If AB and $A'B'$ are any two segments, either $AB > A'B'$, or $AB \cong A'B'$, or $AB < A'B'$, and exactly one of these holds.
7. If $AB > CD$ and $CD \cong EF$, then $AB > EF$.
8. If $AB \cong A'B'$, then $\overline{AB} = \overline{A'B'}$.
9. If $\overline{AB} = \overline{A'B'}$, then $AB \cong A'B'$.
10. $AB > CD$ if and only if $\overline{AB} > \overline{CD}$.
11. The S.A.S. theorem for congruent triangles.
12. The A.S.A. theorem.
13. The S.S.S. theorem.
14. For any two angles $\angle A$ and $\angle B$ either $\angle A < \angle B$, or $\angle A \cong \angle B$, or $\angle A > \angle B$, and exactly one of these is true.
15. All right angles are congruent to one another.
16. An exterior angle of a triangle is greater than either opposite interior angle.

17. In $\triangle ABC$, $AB > AC$ if and only if $\angle C > \angle B$.
18. In $\triangle ABC$, $\overline{AB} + \overline{BC} > \overline{AC}$.
19. If l is any line and A any point not on l, then there exists at least one line on A parallel to l.
20. If parallel lines are cut by a transversal, the alternate interior angles are congruent.
21. The sum of the angles of a triangle is a straight angle.
22. If a transversal is cut by three parallel lines in congruent segments, the same is true for every transversal of these lines.
23. If for $\triangle ABC$ and $\triangle A'B'C'$, $\angle A \cong \angle A'$, $\angle B \cong \angle B'$, and $\angle C \cong \angle C'$, then the triangles are similar.
24. If $\overline{AB}/\overline{A'B'} = \overline{AC}/\overline{A'C'} = \overline{BC}/\overline{B'C'}$, then $\triangle ABC$ is similar to $\triangle A'B'C'$.
25. In a right triangle the altitude on the hypotenuse forms two right triangles similar to the given triangle.
26. In $\triangle ABC$ with $\angle C$ a right angle, $\overline{AC}^2 + \overline{BC}^2 = \overline{AB}^2$.
27. The area of a triangle is one half the base times the altitude.
28. Suppose that circles with centers A and B each have radius $r > \frac{1}{2}\overline{AB}$. Then these circles intersect in two points on opposite sides of the line AB.
29. Exactly one circle is circumscribed about a regular polygon.
30. If regular polygons of n sides are inscribed in two circles, then the perimeters of the polygons are proportional to the radii of the circles.
31. For a given circle let p_n be the perimeter of a regular polygon of n sides inscribed in the circle. The numbers p_n, $n = 3, 4, 5, \ldots$, form a sequence which has a limit.
32. The ratio C/d (circumference to diameter) is the same for all circles.
33. If S_n is the area of a regular polygon of n sides inscribed in a

circle, then limit $\{S_n\}$ exists.

34. The area of a circle is equal to one half the circumference times the radius.
35. If two central angles are congruent, they subtend arcs of equal length.
36. If $\widehat{AB}$ and $\widehat{BC}$ are adjacent arcs (that is, if they have only point B in common), then length $\widehat{ABC}$ = length $\widehat{AB}$ + length $\widehat{BC}$.
37. The measure of an angle inscribed in a circle is one half the measure of the central angle which subtends the same arc.